

第十一周作业

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1. 习题六:

8.

$$(1) \text{ 令 } f(a) = \sum_{i=1}^n (X_i - a)^2$$

$$f'(a) = 2 \sum_{i=1}^n (a - X_i) = 2(na - n\bar{X})$$

令 $f'(a) = 0, a = \bar{X}$, 此时 $f''(a) = 2n > 0, \sum_{i=1}^n (X_i - a)^2$ 最小

$$(2) \sum_{i=1}^n (X_i - \bar{X})^2 = \sum_{i=1}^n (X_i^2 - 2X_i\bar{X} + \bar{X}^2) = \sum_{i=1}^n X_i^2 - 2\bar{X} \sum_{i=1}^n X_i + n\bar{X}^2 = \sum_{i=1}^n X_i^2 - 2n\bar{X}^2 + n\bar{X}^2 = \sum_{i=1}^n X_i^2 - n\bar{X}^2$$

9.

(1) 令 $Y_i = \frac{X_i - \mu}{0.5} = 2X_i$, 则 Y_i 服从标准正态分布

$P(\sum_{i=1}^{10} X_i^2 \geq 4) = P(\sum_{i=1}^{10} Y_i^2 \geq 16), \chi^2 = \sum_{i=1}^{10} Y_i^2$ 服从自由度为 10 的 χ^2 分布

则 $P(\sum_{i=1}^{10} X_i^2 \geq 4) = P(\sum_{i=1}^{10} Y_i^2 \geq 16) \approx 0.1$

$$(2) \frac{1}{0.25} \sum_{i=1}^{10} (X_i - \bar{X})^2 \sim \chi^2(9)$$

故 $P(\sum_{i=1}^{10} (X_i - \bar{X})^2 \geq 4.23) = P(\frac{1}{0.25} \sum_{i=1}^{10} (X_i - \bar{X})^2 \geq 16.92) \approx 0.05$

10.

(1) 由题意得 $X_i \sim N(0, 4)$

故 $\frac{X_1}{2} \sim N(0, 1), \frac{X_1^2}{4} \sim \chi^2(1)$

$X_2 + X_3 + X_4 \sim N(0, 12), \frac{X_2 + X_3 + X_4}{\sqrt{12}} \sim N(1, 0), \frac{(X_2 + X_3 + X_4)^2}{12} \sim \chi^2(1)$

$$\sum_{i=5}^9 \frac{(X_i - Y)^2}{4} \sim \chi^2(4)$$

由卡方分布的可加性,

$$\frac{X_1^2}{4} + \frac{(X_2 + X_3 + X_4)^2}{12} \sim \chi^2(1) + \sum_{i=5}^9 \frac{(X_i - Y)^2}{4} \sim \chi^2(6)$$

故 $a = \frac{1}{4}, b = \frac{1}{12}, c = \frac{1}{4}$

2 (2) 由题意得 $X_1 + X_2 \sim N(0, 8)$, 则 $\frac{X_1+X_2}{\sqrt{8}} \sim N(0, 1)$

$$\frac{X_i}{2} \sim N(0, 1), (\frac{X_3}{2})^2 + (\frac{X_4}{2})^2 + (\frac{X_5}{2})^2 \sim \chi^2(3)$$

$$\text{故 } \frac{\frac{X_1+X_2}{\sqrt{8}}}{\sqrt{\frac{X_3^2+X_4^2+X_5^2}{4 \times 3}}} = \frac{\sqrt{6}}{2} \frac{X_1+X_2}{\sqrt{X_3^2+X_4^2+X_5^2}} \sim t(3)$$

故 $d = \frac{\sqrt{6}}{2}$, 自由度为 3

11.

由题意得 $X_{n+1} \sim N(\mu, \sigma^2)$, $\bar{X} \sim N(\mu, \frac{\sigma^2}{n})$

故 $X_{n+1} - \bar{X} \sim N(0, \frac{n+1}{n}\sigma^2)$, $\frac{X_{n+1}-\bar{X}}{\sqrt{\frac{n+1}{n}\sigma^2}} \sim N(0, 1)$

又因为 $\frac{(n-1)S^2}{\sigma^2} \sim \chi^2(n-1)$

$$\text{故 } Y_1 = \frac{X_{n+1}-\bar{X}}{S} \sqrt{\frac{n}{n+1}} = \frac{\frac{X_{n+1}-\bar{X}}{\sqrt{\frac{n+1}{n}\sigma^2}}}{\sqrt{\frac{(n-1)S^2}{\sigma^2} \frac{n}{n-1}}}, Y_1 \sim t(n-1)$$

因为 $(\frac{X_{n+1}-\bar{X}}{\sqrt{\frac{n+1}{n}\sigma^2}})^2 \sim \chi^2(1)$, $\frac{(n-1)S^2}{\sigma^2} \sim \chi^2(n-1)$

$$\text{故 } Y_2 = \frac{(\frac{X_{n+1}-\bar{X}}{\sqrt{\frac{n+1}{n}\sigma^2}})^2/1}{\frac{(n-1)S^2/\sigma^2}{(n-1)}} = F(1, n-1)$$

12.

由 $X \sim F(12, 12)$, 则 $\frac{1}{X} \sim F(12, 12)$

$$P(X > 1) = P(\frac{1}{X} > 1) = P(X < 1) = P(X \leq 1)$$

又因为 $P(X > 1) + P(X \leq 1) = 1$

故 $P(X > 1) = 0.5$

2. 补充题:

$$\Gamma(1) = \int_0^{+\infty} e^{-t} dt = 1$$

故 $\chi^2(2)$ 的概率密度为

$$f_{\chi^2}(x) = \begin{cases} \frac{1}{2\Gamma(1)} e^{-\frac{x}{2}} = \frac{1}{2} e^{-\frac{x}{2}}, & x > 0 \\ 0, & x \leq 0 \end{cases}$$

$$\text{故 } 2\lambda X \text{ 的分布函数函数为 } P(2\lambda X \leq x) = P(X \leq \frac{x}{2\lambda}) = F_X(\frac{x}{2\lambda}) = \begin{cases} 1 - e^{-\frac{x}{2}}, & x > 0 \\ 0, & x \leq 0 \end{cases}$$

故 $2\lambda X$ 的概率密度为

$$f_{2\lambda X}(x) = F'(x) = \begin{cases} \frac{1}{2}e^{-\frac{x}{2}}, & x > 0 \\ 0, & x \leq 0 \end{cases} = f_{\chi^2}(x)$$

故 $2\lambda X \sim \chi^2(2)$

(2) 由题意得 $2\lambda X_i \sim \chi^2(2)$, 由 χ^2 分布的可加性,

$$2n\lambda\bar{X} = 2n\lambda\frac{1}{n}\sum_{i=1}^n X_i = 2\lambda\sum_{i=1}^n X_i = \sum_{i=1}^n (2\lambda X_i) \sim \chi^2(2n)$$

故 $2n\lambda\bar{X} \sim \chi^2(2n)$

(3) 由于 F 是严格单调递增的连续函数, 故其反函数 F^{-1} 也是严格单调递增的连续函数。

当 $0 < y < 1$ 时, $Y_i = F(X_i)$ 的分布函数为

$$F_{Y_i}(y) = P(F(X_i) \leq y) = P(X_i \leq F^{-1}(y)) = F(F^{-1}(y)) = y$$

当 $y \leq 0$ 时, $F_{Y_i}(y) = 0$; 当 $y \geq 1$ 时, $F_{Y_i}(y) = 1$, 故

$$Y_i \sim U(0, 1)$$

令 $Z_i = -2\ln Y_i$, 当 $z > 0$ 时, Z_i 的分布函数为

$$F_{Z_i}(z) = P(Z_i \leq z) = P(-2\ln Y_i \leq z) = P(Y_i \geq e^{-\frac{z}{2}}) = 1 - P(Y_i < e^{-\frac{z}{2}}) = 1 - F_{Y_i}(e^{-\frac{z}{2}}) = 1 - e^{-\frac{z}{2}}$$

当 $z \leq 0$ 时, $F_{Z_i}(z) = 0$, 故 $Z_i \sim E(0.5)$

由 (1) 知, $Z_i \sim \chi^2(2)$

由 X_i 相互独立, 可知 Z_i 之间相互独立, 则由 χ^2 分布的可加性

$$W = -2\sum_{i=1}^n \ln F(X_i) \sim \chi^2(2n)$$

补充题一:

由题意得 $\bar{X} \sim N(0, \frac{1}{n})$, $\sqrt{n}\bar{X} \sim N(0, 1)$, $n\bar{X}^2 \sim \chi^2(1)$

故 $E(n\bar{X}^2) = nE(\bar{X}^2) = 1$, $D(n\bar{X}^2) = n^2D(\bar{X}^2) = 2$

$$E(\bar{X}^2) = \frac{1}{n}, D(\bar{X}^2) = \frac{2}{n^2}$$

$$\therefore \frac{(n-1)S^2}{1} \sim \chi^2(n-1)$$

$$\therefore E(\frac{S^2}{n}) = \frac{1}{n(n-1)}E((n-1)S^2) = \frac{1}{n}, D(\frac{S^2}{n}) = \frac{1}{n^2(n-1)^2}D((n-1)S^2) = \frac{2}{n^2(n-1)}$$

4 $\therefore \bar{X}$ 与 S^2 相互独立

$$\therefore E(Y) = E(\bar{X}^2) - E\left(\frac{S^2}{n}\right) = 0, D(Y) = D(\bar{X}^2) + D\left(\frac{S^2}{n}\right) = \frac{2}{n(n-1)}$$

补充题二:

由题意得 $\bar{X} \sim N(\mu_1, \frac{\sigma_1^2}{n}), \bar{Y} \sim N(\mu_2, \frac{\sigma_2^2}{m})$

故 $a(\bar{X} - \mu_1) + b(\bar{Y} - \mu_2) \sim N(0, \frac{a^2\sigma_1^2}{n} + \frac{b^2\sigma_2^2}{m})$

$$\therefore \frac{a(\bar{X}-\mu_1)+b(\bar{Y}-\mu_2)}{\sqrt{\frac{a^2\sigma_1^2}{n}+\frac{b^2\sigma_2^2}{m}}} \sim N(0, 1)$$

令 $S_1^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2, S_2^2 = \frac{1}{m-1} \sum_{j=1}^m (Y_j - \bar{Y})^2$, 则

$$\frac{(n-1)S_1^2}{\sigma_1^2} \sim \chi^2(n-1), \frac{(m-1)S_2^2}{\sigma_2^2} \sim \chi^2(m-1)$$

由 χ^2 分布的可加性

$$S^2 = \frac{(n-1)S_1^2}{\sigma_1^2} + \frac{(m-1)S_2^2}{\sigma_2^2} \sim \chi^2(m+n-2)$$

$$\text{故 } \frac{\frac{a(\bar{X}-\mu_1)+b(\bar{Y}-\mu_2)}{\sqrt{\frac{a^2\sigma_1^2}{n}+\frac{b^2\sigma_2^2}{m}}}}{\sqrt{\frac{S^2}{m+n-2}}} = \frac{\sqrt{m+n-2}}{\sqrt{\frac{a^2\sigma_1^2}{n}+\frac{b^2\sigma_2^2}{m}}} \frac{a(\bar{X}-\mu_1)+b(\bar{Y}-\mu_2)}{S} \sim t(m+n-2)$$

$$\text{故 } c = \frac{\sqrt{m+n-2}}{\sqrt{\frac{a^2\sigma_1^2}{n}+\frac{b^2\sigma_2^2}{m}}}, \text{ 自由度为 } m+n-2$$

3. 习题七:

2.

$$E(X) = \int_{-\infty}^{\infty} xf(x; \theta) = \int_0^{\theta} \frac{2\theta^2}{(\theta^2-1)x^2} = \frac{2\theta}{\theta-1}$$

$$\text{故令 } \frac{2\theta}{\theta-1} = \bar{X}$$

解得 θ 的矩估计量为 $\hat{\theta} = \frac{\bar{X}}{2-\bar{X}}$, 矩估计值为 $\hat{\theta} = \frac{\bar{x}}{2-\bar{x}}$